Single Variable Integral Calculus

Indefinite Integration

Standard antiderivatives

- $\displaystyle\int x^n \,\mathrm{d}x = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$
- $\displaystyle\int \frac{1}{x} \,\mathrm{d}x = \ln|x| + C$
- $\displaystyle\int e^x \,\mathrm{d}x = e^x + C$
- $\displaystyle\int \sec^2 x \,\mathrm{d}x = \tan x + C$
- $\displaystyle\int \sec x \tan x \,\mathrm{d}x = \sec x + C$
- $\displaystyle\int \frac{1}{\sqrt{1-x^2}} \,\mathrm{d}x = \sin^{-1} x + C$
- $\displaystyle\int \frac{1}{x\sqrt{x^2-1}} \,\mathrm{d}x = \sec^{-1}|x| + C$
- $\displaystyle\int f^n(x) f'(x) \,\mathrm{d}x = \frac{f^{n+1}(x)}{n+1} + C$
- $\displaystyle\int \tan x \,\mathrm{d}x = -\ln|\cos x| + C$
- $\displaystyle\int a^x \,\mathrm{d}x = \frac{a^x}{\ln a} + C, \quad a > 0,\ a \neq 1$
- $\displaystyle\int \sin x \,\mathrm{d}x = -\cos x + C$
- $\displaystyle\int \cos x \,\mathrm{d}x = \sin x + C$
- $\displaystyle\int \operatorname{cosec}^2 x \,\mathrm{d}x = -\cot x + C$
- $\displaystyle\int \operatorname{cosec} x \cot x \,\mathrm{d}x = -\operatorname{cosec} x + C$
- $\displaystyle\int \frac{1}{1+x^2} \,\mathrm{d}x = \tan^{-1} x + C$
- $\displaystyle\int \frac{f'(x)}{f(x)} \,\mathrm{d}x = \ln|f(x)| + C$
- $\displaystyle\int \cot x \,\mathrm{d}x = \ln|\sin x| + C$
- $\displaystyle\int \operatorname{cosec} x \,\mathrm{d}x = \ln|\operatorname{cosec} x - \cot x| + C$
- $\displaystyle\int \sec x \,\mathrm{d}x = \ln|\sec x + \tan x| + C$
- $\displaystyle\int f(ax+b) \,\mathrm{d}x = \frac{1}{a} F(ax+b) + C, \quad F'(x) = f(x), \quad a \neq 0$
- $\displaystyle\int \sin(ax+b) \,\mathrm{d}x = -\frac{1}{a}\cos(ax+b) + C$
- $\displaystyle\int \cos(ax+b) \,\mathrm{d}x = \frac{1}{a}\sin(ax+b) + C$
- $\displaystyle\int \tan(ax+b) \,\mathrm{d}x = -\frac{1}{a}\ln|\cos(ax+b)| + C$
- $\displaystyle\int \cot(ax+b) \,\mathrm{d}x = \frac{1}{a}\ln|\sin(ax+b)| + C$
- $\displaystyle\int \tan^2 x \,\mathrm{d}x = \tan x - x + C$
- $\displaystyle\int \cot^2 x \,\mathrm{d}x = -\cot x - x + C$
- $\displaystyle\int \sin^2 x \,\mathrm{d}x = \frac{x}{2} - \frac{\sin 2x}{4} + C$
- $\displaystyle\int \cos^2 x \,\mathrm{d}x = \frac{x}{2} + \frac{\sin 2x}{4} + C$
- $\displaystyle\int \sec(ax+b) \,\mathrm{d}x = \frac{1}{a}\ln|\sec(ax+b) + \tan(ax+b)| + C$
- $\displaystyle\int \operatorname{cosec}(ax+b) \,\mathrm{d}x = \frac{1}{a}\ln|\operatorname{cosec}(ax+b) - \cot(ax+b)| + C$
- $\displaystyle\int \sec^3 x \,\mathrm{d}x = \frac{1}{2}\left[\sec x \tan x + \ln|\sec x + \tan x|\right] + C$
- $\displaystyle\int \operatorname{cosec}^3 x \,\mathrm{d}x = \frac{1}{2}\left[-\operatorname{cosec} x \cot x + \ln|\operatorname{cosec} x - \cot x|\right] + C$
- $\displaystyle\int \frac{\mathrm{d}x}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$
- $\displaystyle\int \frac{\mathrm{d}x}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C$

- $\displaystyle\int \frac{\mathrm{d}x}{a^2 - x^2} = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$
- $\displaystyle\int \frac{\mathrm{d}x}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$
- $\displaystyle\int \frac{\mathrm{d}x}{\sqrt{a^2 + x^2}} = \ln \left| x + \sqrt{x^2 + a^2} \right| + C$
- $\displaystyle\int \frac{\mathrm{d}x}{\sqrt{x^2 - a^2}} = \ln \left| x + \sqrt{x^2 - a^2} \right| + C$
- $\displaystyle\int \sqrt{a^2 - x^2}\,\mathrm{d}x = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + C$
- $\displaystyle\int \sqrt{a^2 + x^2}\,\mathrm{d}x = \frac{x}{2}\sqrt{a^2 + x^2} + \frac{a^2}{2}\ln \left| x + \sqrt{x^2 + a^2} \right| + C$
- $\displaystyle\int \sqrt{x^2 - a^2}\,\mathrm{d}x = \frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\ln \left| x + \sqrt{x^2 - a^2} \right| + C$
- $\displaystyle\int \ln|f(x)|\,\mathrm{d}x = x\ln|f(x)| - \int \frac{xf'(x)}{f(x)}\,\mathrm{d}x$
- $\displaystyle\int \ln|x|\,\mathrm{d}x = x\ln|x| - x + C$

Algebraic form integrals

Linear & quadratic

For integrals of the form

- $\displaystyle\int \frac{px+q}{ax^2+bx+c}\,\mathrm{d}x$
- $\displaystyle\int \frac{px+q}{\sqrt{ax^2+bx+c}}\,\mathrm{d}x$
- $\displaystyle\int (px+q)\sqrt{ax^2+bx+c}\,\mathrm{d}x$

Rewrite the numerator as

$px + q = \lambda \dfrac{\mathrm{d}}{\mathrm{d}x}(ax^2 + bx + c) + \mu = \lambda(2ax + b) + \mu.$

Quadratic over Quadratic

For integrals of the form

- $\displaystyle\int \frac{ax^2+bx+c}{px^2+qx+r}\,\mathrm{d}x$
- $\displaystyle\int \frac{ax^2+bx+c}{\sqrt{px^2+qx+r}}\,\mathrm{d}x$

rewrite the numerator as

$$\begin{aligned} ax^2 + bx + c &= \lambda(px^2 + qx + r) + \mu\frac{\mathrm{d}}{\mathrm{d}x}(px^2 + qx + r) + \gamma \\ &= \lambda(px^2 + qx + r) + \mu(2px + q) + \gamma. \end{aligned}$$

Trigonometric form integrals

Sine and Cosine Forms

For integrals of the form

- $\displaystyle\int \frac{\mathrm{d}x}{a\cos^2 x + b\sin^2 x}$
- $\displaystyle\int \frac{\mathrm{d}x}{a + b\sin^2 x}$
- $\displaystyle\int \frac{\mathrm{d}x}{a + b\cos^2 x}$
- $\displaystyle\int \frac{\mathrm{d}x}{(a\sin x + b\cos x)^2}$
- $\displaystyle\int \frac{\mathrm{d}x}{a + b\sin^2 x + c\cos^2 x}$

divide the numerator and denominator by $\cos^2 x$, then use

$$\sec^2 x = 1 + \tan^2 x, \qquad t = \tan x.$$

More Sine and Cosine

For integrals of the form

- $\displaystyle\int \frac{\mathrm{d}x}{a\sin x + b\cos x}$
- $\displaystyle\int \frac{\mathrm{d}x}{a + b\sin x}$
- $\displaystyle\int \frac{\mathrm{d}x}{a + b\cos x}$
- $\displaystyle\int \frac{\mathrm{d}x}{a\sin x + b\cos x + c}$

Use the tangent half-angle substitution

$$\sin x = \frac{2\tan(x/2)}{1 + \tan^2(x/2)}, \qquad \cos x = \frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)}, \qquad t = \tan\frac{x}{2}.$$

Trigonometric numerator and denominator

For integrals of the form

$$\int \frac{p\cos x + q\sin x + r}{a\cos x + b\sin x + c}\,\mathrm{d}x,$$

Rewrite the numerator as

$$\begin{aligned} p\cos x + q\sin x + r = {} & \lambda(a\cos x + b\sin x + c) \\ & + \mu(-a\sin x + b\cos x) + \gamma. \end{aligned}$$

Exponential forms

- $\displaystyle\int e^x\big(f(x) + f'(x)\big)\,\mathrm{d}x = f(x)e^x + C$
- $\displaystyle\int e^{g(x)}\big(f(x)g'(x) + f'(x)\big)\,\mathrm{d}x = f(x)e^{g(x)} + C$
- $\displaystyle\int e^{ax}\sin bx\,\mathrm{d}x = \frac{e^{ax}}{a^2 + b^2}\big(a\sin bx - b\cos bx\big) + C$
- $\displaystyle\int e^{ax}\cos bx\,\mathrm{d}x = \frac{e^{ax}}{a^2 + b^2}\big(a\cos bx + b\sin bx\big) + C$

Integration by Parts

- $\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u$
- $\int u(x)v(x)\,\mathrm{d}x = u(x)\int v(x)\,\mathrm{d}x - \int u'(x)\left(\int v(x)\,\mathrm{d}x\right)\mathrm{d}x$

Reduction Strategies

Powers of sine and cosine

For

$$\int \sin^m x \cos^n x\,\mathrm{d}x, \qquad m, n \in \mathbb{N},$$

use the following cases.

1. If m is odd, retain one factor of $\sin x$, use $\sin^2 x = 1 - \cos^2 x$, and put $t = \cos x$.
2. If n is odd, retain one factor of $\cos x$, use $\cos^2 x = 1 - \sin^2 x$, and put $t = \sin x$.
3. If both m and n are even, use

$$\sin^2 x = \frac{1-\cos 2x}{2}, \qquad \cos^2 x = \frac{1+\cos 2x}{2}.$$

4. If $m, n \in \mathbb{Z}$, $m+n$ is even, and negative powers occur, use one of the following substitutions:

$$t = \tan x,$$
$$\sin^m x \cos^n x\,\mathrm{d}x = t^m(1+t^2)^{-(m+n+2)/2}\,\mathrm{d}t;$$
$$t = \cot x,$$
$$\sin^m x \cos^n x\,\mathrm{d}x = -t^n(1+t^2)^{-(m+n+2)/2}\,\mathrm{d}t.$$

Binomial differentials

For

$$\int x^m(a+bx^n)^p\,\mathrm{d}x, \qquad m, n, p \in \mathbb{Q}, \quad n \neq 0,$$

use the following cases.

1. If $p \in \mathbb{Z}_{\geq 0}$, use the binomial theorem and expand.
2. If $p \in \mathbb{Z}_{<0}$, put $x = t^k$, where k is the least common multiple of the denominators of m and n.
3. If $\dfrac{m+1}{n} \in \mathbb{Z}$ and $p \notin \mathbb{Z}$, put

$$a + bx^n = t^k,$$

where k is the denominator of p in lowest terms.

4. If $\dfrac{m+1}{n} + p \in \mathbb{Z}$ and $p \notin \mathbb{Z}$, put

$$a + bx^n = t^k x^n,$$

where k is the denominator of p in lowest terms.

Definite Integrals

Riemann Sums

- For an integrable function f on $[a,b]$, $\int_a^b f(x)\,\mathrm{d}x = \lim_{n\to\infty} \frac{b-a}{n}\sum_{r=1}^{n} f\left(a + \frac{r(b-a)}{n}\right)$
- For an integrable function f on $[0,1]$, $\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n} f\left(\frac{r}{n}\right) = \int_0^1 f(x)\,\mathrm{d}x$

Basic properties

- $\int_a^a f(x)\,\mathrm{d}x = 0$
- $\int_a^b f(x)\,\mathrm{d}x = -\int_b^a f(x)\,\mathrm{d}x$
- $\int_a^b f(x)\,\mathrm{d}x = \int_a^c f(x)\,\mathrm{d}x + \int_c^b f(x)\,\mathrm{d}x$
- $\int_a^b (\lambda f(x) + \mu g(x))\,\mathrm{d}x = \lambda\int_a^b f(x)\,\mathrm{d}x + \mu\int_a^b g(x)\,\mathrm{d}x$
- If $m \leq f(x) \leq M$ on $[a,b]$, where $a < b$, then $m(b-a) \leq \int_a^b f(x)\,\mathrm{d}x \leq M(b-a)$.

- If $f(x) \leq g(x)$ on $[a, b]$, where $a < b$, then $\displaystyle\int_a^b f(x)\,\mathrm{d}x \leq \int_a^b g(x)\,\mathrm{d}x$.
- If $a < b$, then $\displaystyle\left|\int_a^b f(x)\,\mathrm{d}x\right| \leq \int_a^b |f(x)|\,\mathrm{d}x$

Change of variable and integration by parts

- If f is continuous and ϕ is differentiable, then $\displaystyle\int_a^b f(\phi(x))\phi'(x)\,\mathrm{d}x = \int_{\phi(a)}^{\phi(b)} f(t)\,\mathrm{d}t$
- If u and v are differentiable, then $\displaystyle\int_a^b u(x)v'(x)\,\mathrm{d}x = \big[u(x)v(x)\big]_a^b - \int_a^b u'(x)v(x)\,\mathrm{d}x$

Symmetry properties

- $\displaystyle\int_a^b f(x)\,\mathrm{d}x = \int_a^b f(a+b-x)\,\mathrm{d}x$
- $\displaystyle\int_a^b f(x)\,\mathrm{d}x = \frac{1}{2}\int_a^b [f(x) + f(a+b-x)]\,\mathrm{d}x$
- $\displaystyle\int_0^a f(x)\,\mathrm{d}x = \int_0^a f(a-x)\,\mathrm{d}x$
- $\displaystyle\int_0^{2a} f(x)\,\mathrm{d}x = \int_0^a [f(x) + f(2a-x)]\,\mathrm{d}x$
- If $f(-x) = f(x)$, then $\displaystyle\int_{-a}^a f(x)\,\mathrm{d}x = 2\int_0^a f(x)\,\mathrm{d}x$.
- If $f(-x) = -f(x)$, then $\displaystyle\int_{-a}^a f(x)\,\mathrm{d}x = 0$.
- If $f(a+b-x) = -f(x)$, then $\displaystyle\int_a^b f(x)\,\mathrm{d}x = 0$.
- If $f(a+b-x) = f(x)$, then $\displaystyle\int_a^b xf(x)\,\mathrm{d}x = \frac{a+b}{2}\int_a^b f(x)\,\mathrm{d}x$.
- $\displaystyle\int_a^b \frac{f(x)}{f(x) + f(a+b-x)}\,\mathrm{d}x = \frac{b-a}{2}$, provided the denominator is non-zero.

Periodic and trigonometric properties

- If $T > 0$ and $f(x+T) = f(x)$, then $\displaystyle\int_a^{a+T} f(x)\,\mathrm{d}x = \int_0^T f(x)\,\mathrm{d}x$.
- If $T > 0$, $f(x+T) = f(x)$, and $n \in \mathbb{N}$, then $\displaystyle\int_a^{a+nT} f(x)\,\mathrm{d}x = n\int_0^T f(x)\,\mathrm{d}x$.
- If $T > 0$ and $f(x+T) = -f(x)$, then $\displaystyle\int_a^{a+2T} f(x)\,\mathrm{d}x = 0$.
- $\displaystyle\int_0^{\pi/2} f(\sin x, \cos x)\,\mathrm{d}x = \int_0^{\pi/2} f(\cos x, \sin x)\,\mathrm{d}x$
- $\displaystyle\int_0^{\pi} f(\sin x)\,\mathrm{d}x = 2\int_0^{\pi/2} f(\sin x)\,\mathrm{d}x$
- $\displaystyle\int_0^{\pi} xf(\sin x)\,\mathrm{d}x = \frac{\pi}{2}\int_0^{\pi} f(\sin x)\,\mathrm{d}x$

Wallis formula

- $W_n := \int_0^{\pi/2} \sin^n x \,\mathrm{d}x = \int_0^{\pi/2} \cos^n x \,\mathrm{d}x, \quad n \in \mathbb{Z}_{\geq 0}$
- $W_0 = \dfrac{\pi}{2}, \qquad W_1 = 1$
- $W_n = \dfrac{n-1}{n} W_{n-2}, \quad n \geq 2$
- $W_n W_{n-1} = \dfrac{\pi}{2n}, \quad n \geq 1$
- $W_{2m} = \dfrac{(2m-1)!!}{(2m)!!}\dfrac{\pi}{2} = \dfrac{(2m)!}{2^{2m}(m!)^2}\dfrac{\pi}{2}$
- $W_{2m+1} = \dfrac{(2m)!!}{(2m+1)!!} = \dfrac{2^{2m}(m!)^2}{(2m+1)!}, \quad m \in \mathbb{Z}_{\geq 0}$
- $\int_0^{\pi} \sin^n x \,\mathrm{d}x = 2W_n$
- $\int_0^{2\pi} \sin^n x \,\mathrm{d}x = \int_0^{2\pi} \cos^n x \,\mathrm{d}x = \begin{cases} 4W_n, & n \text{ even}, \\ 0, & n \text{ odd} \end{cases}$
- $W_n = \dfrac{1}{2} B\left(\dfrac{n+1}{2}, \dfrac{1}{2}\right) = \dfrac{\sqrt{\pi}}{2} \dfrac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n+2}{2}\right)}$
- $\dfrac{\pi}{2} = \prod_{k=1}^{\infty} \dfrac{(2k)^2}{(2k-1)(2k+1)}$
- $W_n \sim \sqrt{\dfrac{\pi}{2n}}, \quad n \to \infty$

Gamma function

- $\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} \,\mathrm{d}t, \quad \operatorname{Re} z > 0$
- $\Gamma(z+1) = z\Gamma(z)$
- $\Gamma(1) = 1, \qquad \Gamma(n+1) = n!, \quad n \in \mathbb{Z}_{\geq 0}$
- $\Gamma\left(\dfrac{1}{2}\right) = \sqrt{\pi}$
- $\Gamma\left(n + \dfrac{1}{2}\right) = \dfrac{(2n)!}{4^n n!}\sqrt{\pi} = \dfrac{(2n-1)!!}{2^n}\sqrt{\pi}, \quad n \in \mathbb{Z}_{\geq 0}$
- $\Gamma(z)\Gamma(1-z) = \dfrac{\pi}{\sin(\pi z)}, \quad z \notin \mathbb{Z}$
- $\Gamma(2z) = \dfrac{2^{2z-1}}{\sqrt{\pi}} \Gamma(z)\Gamma\left(z + \dfrac{1}{2}\right)$
- $\int_0^{\infty} x^{s-1} e^{-ax} \,\mathrm{d}x = \dfrac{\Gamma(s)}{a^s}, \quad a, s > 0$
- $\int_0^{\infty} x^m e^{-ax^n} \,\mathrm{d}x = \dfrac{1}{n} a^{-(m+1)/n} \Gamma\left(\dfrac{m+1}{n}\right), \quad a, n > 0,\ m > -1$
- $\int_0^{\infty} e^{-ax^2} \,\mathrm{d}x = \dfrac{1}{2}\sqrt{\dfrac{\pi}{a}}, \qquad \int_{-\infty}^{\infty} e^{-ax^2} \,\mathrm{d}x = \sqrt{\dfrac{\pi}{a}}, \quad a > 0$

Beta function

- $B(p,q) = \int_0^1 t^{p-1}(1-t)^{q-1}\,\mathrm{d}t, \quad p,q>0$
- $B(p,q) = 2\int_0^{\pi/2} \sin^{2p-1} x \cos^{2q-1} x\,\mathrm{d}x$
- $B(p,q) = \int_0^\infty \frac{t^{p-1}}{(1+t)^{p+q}}\,\mathrm{d}t$
- $B(p,q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$
- $B(p,q) = B(q,p)$
- $B(p+1,q) = \frac{p}{p+q}B(p,q), \qquad B(p,q+1) = \frac{q}{p+q}B(p,q)$
- $B(p,1) = \frac{1}{p}$
- $B(m,n) = \frac{(m-1)!(n-1)!}{(m+n-1)!}, \quad m,n\in\mathbb{N}$
- $\int_a^b (x-a)^{p-1}(b-x)^{q-1}\,\mathrm{d}x = (b-a)^{p+q-1}B(p,q), \quad b>a,\ p,q>0$
- $\int_0^{\pi/2} \sin^m x \cos^n x\,\mathrm{d}x = \frac{1}{2}B\left(\frac{m+1}{2},\frac{n+1}{2}\right), \quad m,n>-1$
- $\int_0^\infty \frac{x^{\mu-1}}{(1+x^\nu)^\rho}\,\mathrm{d}x = \frac{1}{\nu}B\left(\frac{\mu}{\nu},\rho-\frac{\mu}{\nu}\right), \quad \nu,\rho>0,\ 0<\mu<\nu\rho$

Basic Theorems of Calculus

Intermediate Value Theorem

- Let $a<b$ and let $f:[a,b]\to\mathbb{R}$ be continuous. If N lies between $f(a)$ and $f(b)$, then there exists $c\in[a,b]$ such that
  $f(c)=N.$

Bolzano's Theorem

- Let $a<b$ and let $f:[a,b]\to\mathbb{R}$ be continuous. If
  $f(a)f(b)<0,$
  then there exists $c\in(a,b)$ such that
  $f(c)=0.$

Extreme Value Theorem

- Let $f:[a,b]\to\mathbb{R}$ be continuous. Then there exist $x_{\min}, x_{\max}\in[a,b]$ such that
  $f(x_{\min}) \le f(x) \le f(x_{\max}) \qquad \text{for every } x\in[a,b].$

Fermat's Theorem

- Let f be defined on an open interval containing c. If f has a local maximum or local minimum at c and is differentiable at c, then
  $f'(c)=0.$

Rolle's Theorem

- Let $a < b$. If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b), and
  $f(a) = f(b)$,
  then there exists $c \in (a, b)$ such that
  $f'(c) = 0$.

Lagrange's Mean Value Theorem

- Let $a < b$. If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ such that
  $$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Cauchy's Mean Value Theorem

- Let $a < b$. If $f, g : [a, b] \to \mathbb{R}$ are continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ such that
  $$[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c).$$

Mean Value Theorem for Integrals

- Let $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous. Then there exists $c \in [a, b]$ such that
  $$\int_a^b f(x)\,\mathrm{d}x = f(c)(b - a).$$
- Let $a < b$, let $f : [a, b] \to \mathbb{R}$ be continuous, and let g be integrable and non-negative on $[a, b]$. Then there exists $c \in [a, b]$ such that
  $$\int_a^b f(x)g(x)\,\mathrm{d}x = f(c)\int_a^b g(x)\,\mathrm{d}x.$$

Fundamental Theorem of Calculus

- Let $a < b$, let $f : [a, b] \to \mathbb{R}$ be continuous, and define
  $$F(x) = \int_a^x f(t)\,\mathrm{d}t.$$
  Then F is continuous on $[a, b]$, differentiable on (a, b), and
  $F'(x) = f(x)$.
- Let $a < b$, let $f : [a, b] \to \mathbb{R}$ be continuous, and let F be any antiderivative of f on $[a, b]$. Then
  $$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

L'Hôpital's Rule

- Let f and g be differentiable on a deleted neighbourhood of a, with $g'(x) \neq 0$. Suppose that either $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$ or $\lim_{x\to a} |f(x)| = \lim_{x\to a} |g(x)| = \infty$. If $\lim_{x\to a} f'(x)/g'(x) = L$, where L is finite or infinite, then
  $$\lim_{x\to a} \frac{f(x)}{g(x)} = L.$$